non-deterministic finite automaton (nfa)

in a dfa, we are always able to determine the current and the next state of the automaton, given an input. in nfa, however, we would not be able to know the state of the automaton until it has finished processing. for example:

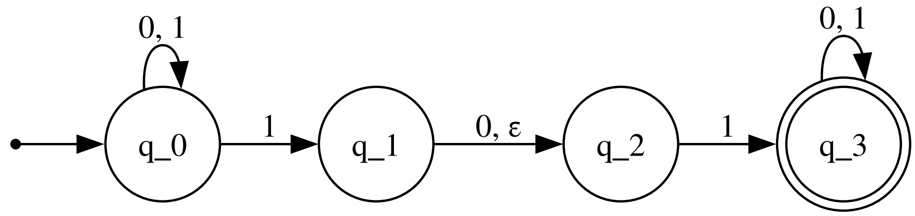

Figure 1: graphs/03_nfa.gv

you might see the difference between an nfa and dfa already. in an nfa, it may have none, one or many exiting arrows for each symbol; comparing to dfa, which *always* has exactly one exiting transition arrow for each symbol in the alphabet.

also in dfa, there are no arrows with the label ϵ. in nfa, however, it may have zero, one, or many arrows labeled ϵ.

to calculate an nfa, we draw a computationl tree; which traces the state of the dfa. take the nfa above for example – we want to compute the machine with the input 010110.

Figure 2: graphs/03_tree.gv